Avrile Floro

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EDUCATION

Institut Polytechnique de Paris Sep. 2025 – Present
PhD Track Master in Data & Artificial Intelligence (jointly offered by École Polytechnique and Télécom Paris)
A selective research-oriented Master's program preparing students for doctoral research. PhD Track supervisor: Prof. Nils Holzenberger (Télécom Paris). Selected coursework: Deep Learning, Collective Intelligence, Database Management Systems, AI Ethics, Logic & Knowledge Representation, Probabilities.

Université Paris 8 Vincennes – Saint-Denis Sep. 2022 – Jul. 2025
Bachelor's Degree in Computer Science, Final grade: 16.7/20, highest honors (mention très bien)
Selected projects: R Shiny app for interactive forecasting of French electricity consumption (rolling forecasts, model performance testing); BERT vs. BART sentiment analysis on a U.S. election subreddit; computer-vision pipeline for invoice information extraction; Self-Organizing Map classifier in C.

Conservatoire National des Arts et Métiers (CNAM) Feb. 2024 – Feb. 2025
Non-degree ECTS coursework in Mathematics and Statistics (30 ECTS)
Linear models, Time-series modeling & forecasting, Statistical techniques, Scientific foundations of mathematics, Mathematical tools for computer science.

Université d'Artois Sep. 2022 – Jul. 2023
Master's Degree in Teaching French as a Foreign, Second, or Specific-Purpose Language, 13.85/20

Université Paris-Dauphine Sep. 2016 – Jul. 2017
Master's Degree in Financial Law (M2), 13.05/20

Duke University, USA Aug. 2015 – May 2016
Master of Laws (LL.M.) with Certificate in Business Law, GPA: 3.226/4

Université Paris-Sud Sep. 2012 – Jul. 2015
Bachelor's Degree in Law (12.49/20); University Diploma in Comparative and International Legal Studies (14.42/20)

RESEARCH EXPERIENCE

Student Researcher, Legal NLP Apr. 2025 – Present
DIG Team, LTCI, Télécom Paris | Supervisor: Prof. Nils Holzenberger
Lead author of *Where Experts Disagree, Models Fail: Detecting Implicit Legal Citations in French Court Decisions* (with T. Dhorasoo, S. Pellez, N. Holzenberger). Available at arXiv:2603.22973. Introduced a benchmark of 1,015 passage-article pairs annotated by three legal experts ($\kappa = 0.33$), and showed that annotator disagreement is the strongest predictor of model failure: 68% of false positives concentrate on the 33% of disputed cases. Designed and implemented the full pipeline: Judilibre data collection, JuriBERT bi-encoder training, adversarial filtering, and supervision of the expert annotation campaign.

Research Intern, NLP and Graph Modeling for French Supreme Court Jurisprudence May 2026 – Aug. 2026
NOS Team, i3 (CNRS), Télécom Paris | Supervisors: Prof. Tiphaine Viard, Prof. Maria Boritchev, Prof. Thomas Le Goff
Computational analysis of a corpus of 548K decisions from the French *Cour de cassation* (Judilibre). Leveraging the highly codified rhetorical structure of *cassation* rulings (visas, motifs, dispositif) to extract legal reasoning patterns capturing jurisprudential evolution over time.

Student Researcher, Multi-Agent Systems & Self-Organisation Mar. 2026 – Present
ACES Team, LTCI, Télécom Paris | Supervisor: Prof. Ada Diaconescu
Project *Collective Nudging that Scales: How Many Dogs Do I Need to Herd Sheep?*: NetLogo agent-based study of how the number of shepherd dogs required to herd a flock scales with flock size. Proposes a local sheep-density gradient as a tractable approximation of the flock's global centre of mass.

Student Researcher, Mechanistic Interpretability of LLMs Feb. 2026 – Present

SAMOVAR, Télécom SudParis | Supervisor: Prof. Luca Benedetto

Project *Cultural Binding Heads in Language Models*: mechanistic investigation of why LLMs default to equal treatment across cultural groups even when context warrants differentiation. Identifies 2–3 mid-layer attention heads causally mediating cultural binding across 8 LLMs, and shows that the bottleneck is routing, not knowledge.

AWARDS & DISTINCTIONS

- **IP Paris PhD Track Excellence Scholarship** (2025–2026, renewable), competitive merit-based scholarship from Institut Polytechnique de Paris: €10,000/year cost-of-living grant for M1 and M2 and subsidized tuition.
- **1st place / 87 teams**, *Influencer or Observer: Predicting Social Roles*, Kaggle competition for the Deep Learning course (INF554) at École Polytechnique, Sept.–Dec. 2025. Built an NLP classifier predicting users' social roles (Influencer vs. Observer) from tweet content alone.

TEACHING EXPERIENCE

Academic Tutor, Computer Science Bachelor's Program Nov. 2024 – Jul. 2026

Université Paris 8 | 218 hours/year

Provide academic and methodological support to undergraduate Computer Science students; liaise between teaching staff and students.

French Teacher Aug. 2020 – Present

The French Class, San Francisco, USA

One-on-one and group classes from A1 to C2; weekly design of personalized teaching materials; developed an automated note-organization tool.

TECHNICAL SKILLS & LANGUAGES

Programming: Python (proficient), R, C, Java, Lisp
ML / NLP stack: PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy
Other tools: R Shiny, NetLogo, Git, LaTeX

Languages: French (native), English (C1–C2), Spanish (B2)